SPARKY'S SCIENCE FACTS

What Are Fractions?

Mathematics · Ages 7+

> **Short answer:** A fraction is a part of a whole that has been cut into equal parts. A half means 1 of 2 equal parts.

How it works

- The bottom number, the denominator, says how many equal parts the whole is cut into.
- The top number, the numerator, says how many of those parts you have. Three quarters is 3 of 4 equal parts.
- The parts must be equal: a bread cut into one big and one small piece is not cut in half.
- Different fractions can be the same amount: one half is the same as two quarters and three sixths.
- To add fractions with the same bottom number, add the top numbers: one quarter plus two quarters is three quarters.

Key words: Numerator: the top number; Denominator: the bottom number

Common mix-ups

- One eighth is smaller than one quarter, even though 8 is bigger than 4. More parts means smaller pieces.
- When adding, do not add the bottom numbers: one quarter plus one quarter is two quarters, not two eighths.

Did you know? 50 cents is half of R1, and 25 cents is a quarter of R1.

Try it: Fold a piece of paper in half, then in half again. Open it: you have 4 quarters!

Ask Sparky: Is one third bigger than one half? How do I share 3 pizzas between 4 people?